FORMOSA CRYPTO

Completing the Chain:

Verified Implementations of Hash-Based Signatures and Their Security

Manuel Barbosa François Dupressoir **Rui Fernandes**
Andreas Hülsing Matthias Meijers Pierre-Yves Strub

Repository

Paper

CHES 2026

TCHES 2026(3), pp. 958–981

Jasmin [ABB+17, ABB+20]

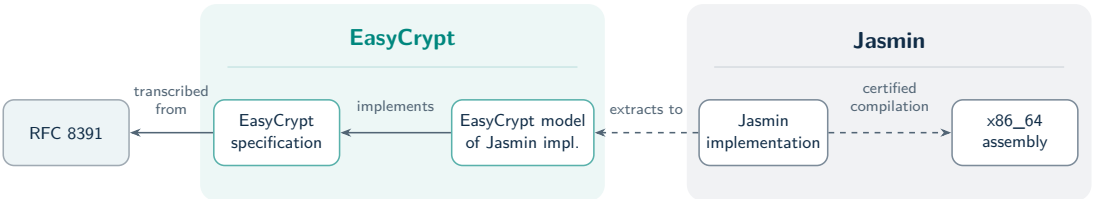
Jasmin [ABB+17, ABB+20]

- Stack zeroization [OBG+24]
- (Static) Safety checker based on abstract interpretation
- Compiler correctness proved in Rocq
- Extraction to EasyCrypt

Compiler Correctness & Constant-Time Preservation

Compilation preserves the semantics of *safe* source programs and constant-time behavior: functional correctness and constant-time guarantees established at source level carry to the assembly level.

Verification Workflow



Trusted Computing Base

- Correctness of SHA2 is assumed

```
axiom hash_128 (x : W8.t Array128.t) :
  phoare[
    M(Syscall).hash_plen_3n____sha256 :
    arg.`2 = x
    ==>
    to_list res = NBytes.val (Hash (to_list x))
  ] = 1%r.
```

Figure: EasyCrypt axiom stating the correctness of SHA2.

QUESTIONS?

References I

[ABB+17] José Bacelar Almeida, Manuel Barbosa, Gilles Barthe, Arthur Blot, Benjamin Grégoire, Vincent Laporte, Tiago Oliveira, Hugo Pacheco, Benedikt Schmidt, and Pierre-Yves Strub.

Jasmin: High-assurance and high-speed cryptography.

In Bhavani M. Thuraisingham, David Evans, Tal Malkin, and Dongyan Xu, editors, *ACM CCS 2017*, pages 1807–1823. ACM Press, October / November 2017.

[ABB+20] José Bacelar Almeida, Manuel Barbosa, Gilles Barthe, Benjamin Grégoire, Adrien Koutsos, Vincent Laporte, Tiago Oliveira, and Pierre-Yves Strub.

The last mile: High-assurance and high-speed cryptographic implementations.

In *2020 IEEE Symposium on Security and Privacy*, pages 965–982. IEEE Computer Society Press, May 2020.

[OBG+24] Santiago Arranz Olmos, Gilles Barthe, Ruben Gonzalez, Benjamin Grégoire, Vincent Laporte, Jean-Christophe Léchenet, Tiago Oliveira, and Peter Schwabe.

High-assurance zeroization.

IACR TCHES, 2024(1):375–397, 2024.